

Greening Nigeria's Cities: A Case Study on Renewable Energy in Sustainable Urban Development

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ABSTRACT

This study examines how renewable energy can be used in sustainable building projects in three major Nigerian cities: Lagos, Abuja, and Port Harcourt. It looks at the potential of solar panels (PV systems), wind energy, and converting biomass waste into energy to reduce both energy use and greenhouse gas emissions. To gather insights, the research involved interviews, site visits, and computer simulations using energy modelling software. The analysis included cost-benefit studies, sensitivity checks, and life cycle assessments to measure both economic and environmental outcomes. Results show that solar PV systems are the most efficient, cutting energy use by 25% to 35% and paying for themselves in 6 to 8 years. Biomass energy showed good potential in industrial areas, while wind energy is less common due to high upfront costs and slower returns. The study highlights key challenges such as high starting costs, limited technical skills, and weak enforcement of energy policies. To overcome these barriers, the paper suggests increasing financial support, strengthening policy implementation, investing in local skills development, and encouraging public-private partnerships to promote the use of renewable energy in Nigerian cities.

1. INTRODUCTION

The construction sector plays a major role in climate change, contributing around 39% of energy-related carbon dioxide emissions globally [1]. As climate concerns grow, the push for sustainable building practices has intensified. These practices aim to reduce resource use and environmental harm throughout a building's life span. A key part of this effort is adding renewable energy (RE) solutions to building designs, which helps cut down on fossil fuel use and supports long-term energy efficiency [2]. Technologies such as solar power, wind energy, biomass, and small-scale hydropower can help buildings lower their carbon emissions while providing more stable and cleaner energy sources.

In Nigeria, finding sustainable energy solutions is especially urgent. Cities like Lagos, Abuja, and Port Harcourt are expanding rapidly, which puts more pressure on already struggling energy systems. The national power grid is often unreliable and inefficient, leading many homes and businesses to depend on diesel generators that are both costly and polluting [3]. This energy gap affects economic progress and worsens environmental problems.

Using renewable energy in urban building projects can help ease these challenges. However, despite Nigeria's rich solar and wind resources, renewable energy use in construction remains limited due to financial, technical, and policy issues [4].

Nigeria is also facing serious environmental threats due to climate change. Rising temperatures, irregular rainfall, and extreme weather are becoming more common, which increases the need for cooling in cities and raises energy demand [5]. At the same time, the country's heavy reliance on fossil fuels has made it one of the largest greenhouse gas emitters in sub-Saharan Africa, contributing to poor air quality and global warming [6]. Shifting urban energy systems toward renewables is therefore crucial for reducing emissions and improving environmental sustainability.

From an economic point of view, the impact of unreliable energy is also significant. Frequent power outages cost Nigeria an estimated \$28 billion every year in lost productivity [7]. High electricity costs and unstable supply hurt both households and businesses. Renewable energy provides a promising alternative—it can lower energy expenses over time, reduce dependence on the



national grid, and open up new economic opportunities. For example, investing in solar panel production and wind energy installation could create jobs and support urban economic growth [8].

This study focuses on the need to expand the use of renewable energy in sustainable building projects across Nigeria's urban centers. It explores how technologies like solar, wind, and biomass are currently being used, how effective they are in cutting energy use and emissions, and the main barriers to their wider adoption. The study also looks at the broader social and economic benefits of renewable energy and offers practical recommendations for overcoming existing challenges.

2. LITERATURE REVIEW

Understanding Sustainable Buildings and Renewable Energy

Sustainable buildings, also known as green buildings, aim to reduce their environmental impact across their entire life cycle. They focus on improving energy and water efficiency, reducing waste, and using environmentally friendly materials [9]. One critical strategy for achieving these goals is the integration of renewable energy (RE) solutions—energy generated from naturally replenished sources such as sunlight, wind, biomass, and hydropower [10].

The combination of sustainable building practices with renewable energy offers major benefits, including long-term cost savings, lower carbon emissions, and improved energy security [11]. Since buildings are responsible for about 28% of global energy-related CO₂ emissions, technologies such as solar PV panels, wind turbines, and biomass heating systems are increasingly being used to cut emissions and promote clean energy in the built environment [12]. In addition, renewable systems allow for decentralized energy generation, easing the burden on national grids and enhancing resilience during outages [13].

Global Practices and Emerging Trends

Around the world, countries are prioritizing renewable energy in buildings to meet climate targets and transition to low-carbon economies. In the European Union, the Energy Performance of Buildings Directive (EPBD) has driven adoption by requiring all new buildings to be nearly zero-energy by 2021 [14]. Germany's "Energiewende" policy is a leading example, combining financial incentives with strong policy support to expand solar PV usage in both residential and commercial buildings. By 2020, over 40% of energy used in Germany's building sector came from renewable sources [15, 16].

In the United States, programs like the LEED certification and federal tax incentives have encouraged renewable energy use in buildings. States with high solar potential, such as California and Texas, have embraced these technologies. In California, for example, solar energy accounted for nearly 25% of the electricity used in new

homes as of 2022, reflecting a major shift toward clean energy integration in the housing sector [17, 18].

Developing countries have progressed more slowly due to financial, technical, and regulatory challenges [19]. However, India has made significant strides with rooftop solar installations supported by the National Solar Mission, targeting 40 GW of rooftop capacity by 2022 [20]. Similarly, South Africa's Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) has helped promote large-scale renewable energy projects, although the adoption of RE at the building level remains limited [21].

Sustainable Building in Developing Countries

In many developing nations, sustainable building practices are gaining attention as a way to tackle energy shortages and the environmental impact of rapid urbanization. These countries face unique pressures, such as population growth, limited infrastructure, and energy supply gaps, which make clean, efficient urban development critical [22]. In Nigeria, urban expansion is driving up electricity demand, placing additional strain on an already unreliable national grid [23].

Despite these challenges, progress has been made elsewhere on the continent. Kenya has implemented a policy requiring solar water heating systems in new buildings, which has significantly boosted the use of solar technologies in cities [24]. Rwanda has taken steps to encourage energy-efficient construction and the use of renewables, especially in its rapidly growing capital, Kigali [25]. These examples demonstrate that with the right policies and support, sustainable building and renewable energy integration are achievable in African urban centers.

Nigeria has strong renewable energy potential, especially in solar and biomass. The country receives an average of 5.5 kWh/m² of solar radiation daily, making solar PV systems highly viable for urban energy needs [26]. In addition, the large volume of agricultural waste generated across Nigeria could serve as a valuable source of biomass energy [27]. However, this potential has not yet been fully tapped due to a range of barriers that are discussed further in this study.

Renewable Energy Options for Nigerian Cities

Several renewable energy technologies are suitable for use in Nigerian urban buildings, each with its own strengths and limitations. Solar power stands out as the most practical solution, thanks to Nigeria's abundant sunlight. Solar photovoltaic (PV) systems can be easily installed on rooftops and used for both residential and commercial electricity generation. The falling cost of PV technology has improved affordability, although the initial investment still presents a hurdle for many property developers [28]. Recent advances in battery storage technology have made solar power more reliable by allowing energy to be stored for use during low-sunlight periods, which is especially valuable for urban energy users [29].

Wind energy has seen limited adoption in Nigeria because average wind speeds are generally low. However, some coastal and northern regions have shown promise for small-scale wind energy development. Wind turbines could be used in combination with solar systems in such areas, though their use remains constrained by geographic and climatic factors [30].

Biomass energy, derived from organic materials such as agricultural waste and animal manure, offers another renewable option. It can be used for electricity, heating, and cooking. In urban settings, biomass energy could be incorporated into building designs through systems like biogas digesters and biomass boilers. Nonetheless, logistical issues—such as the difficulty of collecting and transporting feedstock—have limited its widespread use in cities [31].

Small-scale hydropower also presents opportunities, especially in areas with access to rivers or flowing water. These systems can provide a stable power source for certain communities. However, hydropower is better suited to rural locations than dense urban areas, where space and water access are limited [32].

Barriers to Renewable Energy Adoption in Nigeria

The slow adoption of renewable energy in Nigerian cities is largely due to several financial, technical, and regulatory challenges. One of the biggest obstacles is the high upfront cost of installing systems like solar PV. Although these technologies are becoming cheaper globally, the capital investment required remains unaffordable for many Nigerian developers and property owners, especially in lower-income areas [33]. Moreover, limited access to credit and the absence of low-interest loans have made it difficult to finance renewable projects. Many banks and financial institutions view these projects as risky, which discourages investment [34].

Technical constraints also hinder progress. There is a notable shortage of trained technicians capable of installing and maintaining renewable systems, particularly for complex technologies like solar PV and wind turbines [35]. This shortage drives up costs and reduces the long-term sustainability of installations. Additionally, most renewable components are imported, as Nigeria has little domestic manufacturing capacity for solar panels or wind turbines. This reliance on imports raises costs and causes delays in project delivery [36].

From a policy perspective, Nigeria has introduced various strategies to encourage renewable energy adoption, including the Renewable Energy Master Plan and the National Renewable Energy and Energy Efficiency Policy. However, enforcement remains weak, and bureaucratic inefficiencies have slowed implementation. Developers also lack strong incentives, such as tax breaks or subsidies, that would make RE adoption more attractive and financially viable [37].

Policy Landscape and Recommendations

Nigeria's policy framework has taken steps to promote renewable energy in buildings, but results have been mixed. The Renewable Energy Master Plan (REMP), led by the Energy Commission of Nigeria, aims to expand the country's renewable energy capacity to 13,000 MW by 2030 [38]. The National Renewable Energy and Energy Efficiency Policy (NREEEP) also supports the use of renewable systems and energy-saving materials in new construction projects [39]. However, both policies have struggled with enforcement, and without incentives, their real-world impact has been limited.

To improve adoption, several recommendations have been proposed. These include introducing tax relief and subsidies for renewable projects, building partnerships between the public and private sectors to fund installations, and investing in local training programs to grow technical expertise [40]. Implementing these measures could help overcome the current financial and technical barriers and move Nigeria closer to a more sustainable, energy-efficient urban future.

3. RESEARCH METHODOLOGY

Case Study Design and Site Selection

This research adopted a case study approach to examine how renewable energy is being integrated into sustainable building projects in Nigeria. This method was chosen because it allows for in-depth analysis of complex, real-world issues in specific contexts [41]. The study focused on three major urban centers—Lagos, Abuja, and Port Harcourt—selected for their rapid urban growth, high energy needs, and renewable energy potential. Lagos, Nigeria's commercial hub, has intense energy consumption and an urgent need for energy alternatives. Abuja, the capital, is undergoing major infrastructure development, making it an ideal setting for studying sustainable urban planning. Port Harcourt, the core of the oil and gas industry, was included to examine renewable integration in an industrialized urban environment [42].

Geographical and climatic conditions were also factored into the selection. Lagos, located on the coast, receives high solar radiation and has offshore wind energy potential. Abuja, situated centrally, benefits from consistent solar radiation and access to biomass from agricultural activity. Port Harcourt, despite its high humidity, also shows solar potential. These varied conditions provided a solid basis for comparing renewable energy solutions across different urban settings [43].

Data Collection

A mixed-methods approach was used to gather both primary and secondary data on renewable energy integration. Primary data came from 30 semi-structured interviews conducted across the three cities. Interviewees included architects, engineers, urban planners, developers, and government officials involved in renewable energy and sustainable urban development. These interviews

explored their experience with renewable technologies, perceived challenges, and opinions on scalability and feasibility [44].

Site visits to selected buildings with renewable energy systems were also conducted to gather direct observations and technical data, including system performance and maintenance practices. Secondary data were sourced from official policy documents, academic publications, and government reports, such as the Nigerian Renewable Energy Master Plan (REMP) and the National Renewable Energy and Energy Efficiency Policy (NREEEP) [45].

Data Analysis

Qualitative data from the interviews were analyzed using thematic analysis to identify recurring issues related to the adoption of renewable energy in buildings. This method was suitable for uncovering stakeholder perspectives on policy, financial, and technical challenges [46]. NVivo software was used to code and categorize the interview transcripts, making it easier to extract key themes [47].

For quantitative analysis, energy performance simulations were conducted using EnergyPlus software. This tool modeled energy use in buildings with and without renewable energy systems under the specific climatic conditions of Lagos, Abuja, and Port Harcourt. The simulations assessed energy savings (in kWh), reductions in greenhouse gas (GHG) emissions (in metric tons of CO₂), and potential cost savings from reduced utility bills [48].

Financial and Environmental Impact Analysis

To assess the financial feasibility of integrating renewable systems, a cost-benefit analysis (CBA) was conducted. This analysis calculated the net present value (NPV) and internal rate of return (IRR) for various technologies, including solar PV, biomass digesters, and wind turbines. Projections were made over a 20-year lifespan, factoring in installation costs, savings from reduced grid use, and long-term maintenance expenses [49].

A sensitivity analysis was carried out to test how changes in factors such as equipment costs or energy prices would affect financial outcomes. This helped to assess the robustness of investment returns under different policy or market conditions [50]. A break-even analysis was also included to determine how long it would take for building owners to recover their initial investment in each renewable energy system [51].

Environmental impacts were assessed through a life cycle assessment (LCA), which measured emissions and energy use from the manufacturing, installation, operation, and end-of-life phases of renewable energy systems. Key indicators included energy payback periods and total GHG reductions over the life span of each system [52]. This provided a comprehensive view of both financial and ecological benefits.

Study Limitations

Despite its comprehensive approach, the study faced several limitations. Relying on interviews as a primary data source introduced the risk of bias, as responses were based on personal opinions and experiences. To improve validity, these findings were cross-verified with secondary sources, including official reports and technical literature [53]. Site visits were informative, but their limited number meant that results may not be fully generalizable to all renewable projects in the selected cities.

Financial estimates in the CBA were based on projections and similar international case studies due to the lack of detailed financial records for local projects. While these references helped fill data gaps, they may not fully reflect the Nigerian market context. Likewise, the simulations used to estimate energy savings and GHG reductions were based on standard performance models and may not capture real-world deviations caused by local maintenance practices, system degradation, or user behavior [54].

Future research should focus on gathering long-term operational data from renewable energy projects currently in use across Nigerian cities. This would improve the accuracy of financial and environmental evaluations. As technologies and policies continue to evolve, ongoing monitoring will be crucial for understanding how renewable energy can more effectively support sustainable urban development in Nigeria.

4. RESULTS AND DISCUSSION

Energy Landscape of Selected Urban Centers

The three cities studied—Lagos, Abuja, and Port Harcourt—show different patterns in energy demand, renewable energy potential, and urban development. Lagos, as Nigeria's economic center, consumes about 40% of the country's total electricity [55]. Due to this high demand, solar energy has proven to be the most practical renewable option for both homes and businesses. With an average solar radiation of 5.5 kWh/m²/day, Lagos is well-suited for rooftop solar PV installations, which can generate significant energy savings.

Abuja, the nation's capital, is expanding quickly, especially in terms of residential construction. As energy needs grow, government incentives have encouraged the adoption of solar PV systems, particularly in new housing estates. This has positioned Abuja as one of the leading cities in Nigeria for integrating solar energy into urban development [56].

Port Harcourt, heavily reliant on the oil and gas sector, often experiences power outages despite being an energy-producing city. While fossil fuels still dominate, interest in alternative energy is increasing. Biomass waste-to-energy systems have shown promising results in industrial settings, while residential use of solar energy is gradually expanding [57].

Adoption of Renewable Energy in Buildings

Although still in the early stages, the use of renewable energy technologies in buildings is increasing across all three cities, with solar PV systems being the most commonly used. In Lagos, many commercial buildings have adopted rooftop solar PV systems to reduce their dependence on diesel generators. One example is a commercial facility that saved approximately 45,000 kWh of electricity in one year—cutting its use of grid power by 25% [58].

In Abuja, solar PV systems have become more common in residential communities. Some housing estates have reported annual energy savings of up to 35,000 kWh, with overall electricity usage reduced by about 30% [59].

Port Harcourt has also made progress. A municipal facility in the city implemented a waste-to-energy system that converts organic waste into biogas for electricity

generation. This project has reduced its grid electricity use by 20%, producing about 10,000 kWh annually [60]. While solar use is growing in homes, high installation costs continue to slow its widespread adoption.

Summary of Case Study Results

The case studies (Figure 1) revealed varying levels of success in adopting renewable technologies across the three cities. In Lagos, a large commercial solar PV project led to a 25% drop in grid electricity use, with yearly savings of 45,000 kWh. Abuja's example of a hybrid solar-diesel residential system reduced reliance on the grid by 30%, saving about 35,000 kWh annually [61]. In Port Harcourt, a biomass waste-to-energy system used in a public facility generated 10,000 kWh per year and cut grid energy usage by 20%, highlighting the viability of biomass energy in industrial and institutional settings [62].

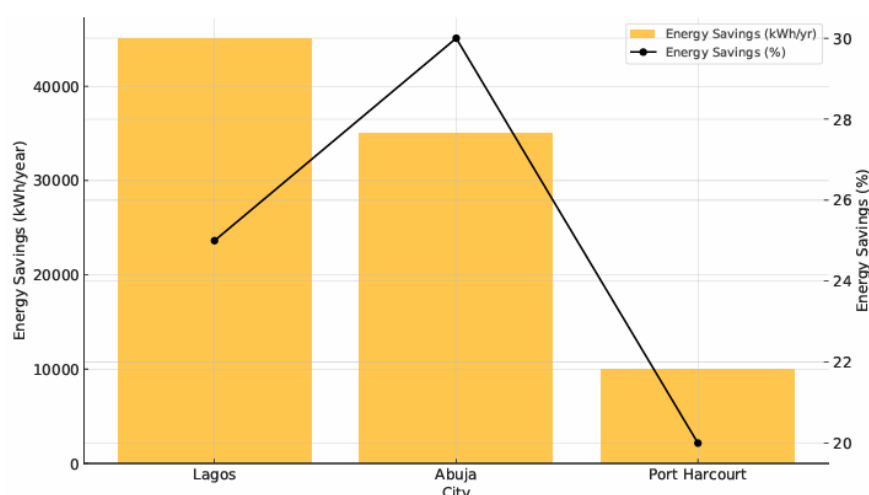


Figure 1. Annual Energy Savings by Renewable Energy Solutions in Nigerian Urban Centers

The study shows that solar PV systems offer the highest energy savings among the renewable technologies evaluated, particularly in Lagos and Abuja. In Lagos, commercial solar PV installations achieved annual savings of 45,000 kWh (25%), while in Abuja, residential estates saved 35,000 kWh (30%) annually. In contrast, Port Harcourt's waste-to-energy system, though promising, yielded a more modest 10,000 kWh per year, reducing grid reliance by 20%. Lagos and Abuja outperformed Port Harcourt due to stronger solar adoption, higher energy demands, and more supportive infrastructure. Both cities benefit from strong solar radiation, but Port Harcourt's coastal climate—with higher humidity and cloud cover—reduces solar efficiency. Moreover, renewable energy projects in Lagos and Abuja are often better funded, supported by government incentives, and located in buildings with larger rooftop space or higher daily energy consumption.

Comparing Lagos and Abuja, Lagos recorded higher total energy savings due to large-scale commercial use, while Abuja achieved a higher percentage reduction because of efficient hybrid systems in residential estates.

These results are based on direct observations from site visits, stakeholder interviews, and data modelling. They are further supported by policy reports such as the Renewable Energy Master Plan and NREEEP. Overall, Lagos and Abuja are leading in solar integration, while Port Harcourt shows early potential for industrial-scale biomass applications.

Comparative Analysis of Renewable Energy Technologies

A comparative analysis was conducted (see Table 1) to assess how solar PV, wind energy, and biomass technologies perform in terms of energy savings, cost-effectiveness, and environmental benefits. Among these, solar photovoltaic (PV) systems delivered the highest energy savings in all three cities, with reductions ranging from 25% to 35%. These results varied depending on the size and efficiency of each installation. Wind energy showed some promise, particularly in areas with stronger wind conditions—such as coastal regions near Lagos and parts of northern Nigeria. Projects in these locations achieved energy savings between 15% and 20%. However, the adoption of wind energy remains low due to its high

initial costs and the need for advanced technical equipment and maintenance expertise [59].

Biomass energy, particularly through waste-to-energy systems, also demonstrated strong potential. In Port Harcourt, such systems provided moderate energy savings of 20% to 30%, but their greatest impact was environmental. By converting organic waste into biogas, these systems significantly reduced greenhouse gas emissions, making them a valuable option for industrial and public infrastructure applications [63].

Figure 2 presents a comparative analysis of estimated annual energy savings from three renewable energy technologies—solar photovoltaic (PV), wind energy, and biomass—across the three case study cities: Lagos, Abuja, and Port Harcourt. The values, expressed in kilowatt-hours per year (kWh/yr), are based on actual case study data, supported by secondary estimates derived from system performance benchmarks, stakeholder interviews, and site observations.

Lagos demonstrated the highest energy savings from solar PV systems, with an estimated 45,000 kWh/yr. This figure is based on a commercial-scale installation that achieved a 25% reduction in grid electricity consumption [58]. Wind energy savings in Lagos were estimated at 30,000 kWh/yr, reflecting potential capacity in coastal regions with moderate wind speeds [59]. Biomass systems, though less prevalent, were estimated to provide annual savings of approximately 20,000 kWh, assuming moderate adoption in industrial zones where organic waste is readily available [63].

In Abuja, solar PV also led in energy savings with an estimated 35,000 kWh/yr. This value is drawn from housing estates that have integrated solar PV systems into their infrastructure, achieving a 30% reduction in electricity use [59]. Wind energy was estimated to contribute 25,000 kWh/yr, based on regional wind resource availability and projected system efficiency [59]. Biomass systems, supported by agricultural activity surrounding the city, were estimated to provide savings of 15,000 kWh/yr, though large-scale implementation remains limited.

Port Harcourt, with its industrial profile and reliance on fossil fuels, showed lower but noteworthy levels of renewable energy adoption. A municipal waste-to-energy facility accounted for approximately 10,000 kWh/yr in energy savings, demonstrating the environmental and energy potential of biomass in public infrastructure [60], [62]. Solar PV and wind systems were estimated to contribute 10,000 kWh and 15,000 kWh annually,

respectively, based on current residential use patterns and geographic feasibility.

These figures highlight the dominance of solar PV systems in urban energy strategies, with consistent performance and broader adoption across all three cities. Wind energy remains underutilized due to its higher upfront costs and infrastructure requirements, though its potential is evident in Lagos and Abuja. Biomass offers significant environmental benefits, particularly through waste repurposing, though logistical and policy challenges have limited its scale.

Energy Savings and Environmental Benefits

Simulations carried out using EnergyPlus software revealed significant energy and environmental gains from integrating renewable energy into buildings in Nigerian cities. In Lagos, a commercial building fitted with a solar PV system reduced its yearly energy use by 25%, which amounted to savings of around 50,000 kWh. This energy reduction also led to a decrease of approximately 50 tons of carbon dioxide emissions annually, demonstrating the strong environmental advantage of solar power [64].

In Abuja, residential estates using hybrid systems that combine solar and diesel power achieved a 30% cut in their reliance on the national grid. These systems saved about 35,000 kWh of energy per year and reduced carbon emissions by 35 tons annually [61]. Port Harcourt's waste-to-energy project also showed measurable benefits. By converting organic waste into biogas for electricity, the system reduced grid electricity use by 20%, generating 10,000 kWh annually. This resulted in an estimated reduction of 20 tons of carbon dioxide emissions per year [62].

Figure 3 illustrates that solar energy contributed the highest CO₂ emissions reductions in Lagos and Abuja, underscoring its strong environmental benefits in these cities. Lagos and Abuja benefit from high solar radiation levels, widespread adoption of rooftop PV systems, and supportive government policies. These factors have enabled solar installations to perform efficiently, with systems in both cities achieving substantial annual CO₂ reductions—45 tons in Lagos and 35 tons in Abuja. The energy payback period for solar PV systems in these regions is also favourable, ranging between two and three years, which enhances their overall sustainability and environmental impact [65, 66].

Table 1. Comparative Analysis of Renewable Energy Solutions in Nigerian Urban Centers

Renewable Energy Source	Energy Savings (%)	Capital Cost (USD/kW)	Payback Period (Years)	GHG Emissions Reduction (tons of CO ₂ /year)
Solar Photovoltaic (PV)	25 – 35	1,200 – 1,800	6 – 8	50 – 70
Wind Energy (Small-scale)	15 – 20	2,000 – 2,500	10 – 12	30 – 50
Biomass (Waste-to-Energy)	20 – 30	1,800 – 2,200	8 – 10	40 – 60

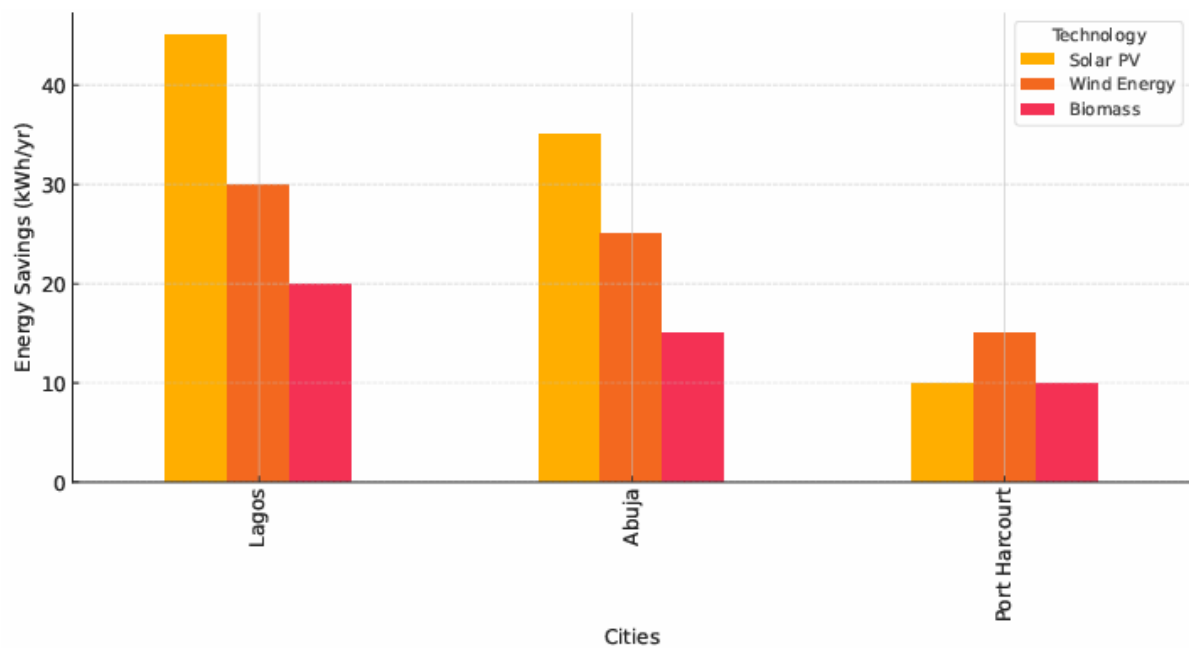


Figure 2. Energy Savings by Renewable Energy Technology in Lagos, Abuja, and Port Harcourt

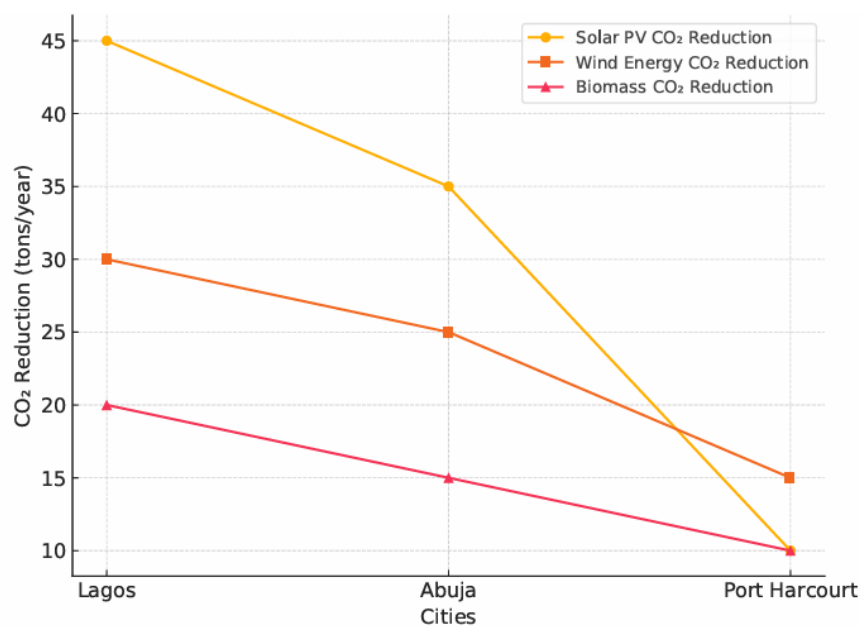


Figure 3. Annual CO₂ Emissions Reductions from Renewable Energy Solutions

In contrast, solar energy in Port Harcourt resulted in lower emissions reductions, with only 10 tons of CO₂ saved annually. This outcome is due to several local challenges. The city's humid, cloudy climate reduces solar panel efficiency, and its energy landscape is dominated by fossil fuel-driven industrial activities, where solar integration remains limited. Additionally, there is less policy support and infrastructure development for solar adoption compared to Lagos and Abuja. Interestingly, biomass waste-to-energy systems in Port Harcourt proved more effective than solar in emissions reduction, particularly in

industrial and public facilities. These systems help mitigate methane emissions from organic waste, contributing to a notable 20-ton CO₂ reduction annually despite offering lower direct energy savings [66]. This highlights the importance of selecting renewable solutions that align with each city's environmental conditions, energy demand profile, and resource availability.

Cost-Benefit Analysis (CBA)

The cost-benefit analysis (CBA), as shown in Table 2, evaluated the financial viability of different renewable

energy systems by comparing their installation costs with long-term savings from reduced electricity bills and lower dependence on the national grid. In Lagos and Abuja, solar photovoltaic (PV) systems showed a positive net present value (NPV) and a payback period of 6 to 8 years, confirming that they are economically viable. These results are consistent with global trends, where solar PV systems have shown competitive financial performance in similar urban environments [66, 67].

In Port Harcourt, biomass-based waste-to-energy systems also returned a positive NPV with a slightly longer payback period of 8 to 10 years. This is largely due to the availability of low-cost organic waste and the added environmental benefit of reducing methane emissions from landfills. On the other hand, small-scale wind energy systems were found to be less financially appealing. With a payback period of 10 to 12 years and higher capital costs, they produced a neutral NPV. These findings suggest that wind energy projects in Nigeria would require policy incentives or subsidies to become economically attractive [68].

These findings highlight that solar PV and biomass are the most financially viable renewable energy options for Nigeria's urban buildings. Meanwhile, wind energy remains less competitive without additional financial support. Similar studies in other developing countries have drawn the same conclusion, reinforcing the importance of government policies to improve the investment outlook for wind technologies [69].

Sensitivity Analysis

A sensitivity analysis was carried out to examine how changes in key variables—such as equipment costs and government subsidies—affect the financial viability of renewable energy investments. The aim was to assess how resilient each technology is under shifting economic or policy conditions. The results are presented in Table 3. In Lagos and Abuja, solar PV systems remained financially viable even with a 10% rise in equipment costs. This resilience is largely due to the ongoing global decline in the cost of solar technology, which helps offset minor increases in local installation expenses [70].

By contrast, wind energy systems were found to be much more vulnerable to cost changes. A 10% increase in equipment costs or a 10% reduction in government subsidies had a significant negative impact on the financial returns of wind projects. This suggests that wind energy in Nigeria is highly dependent on policy support, including subsidies and incentives, to remain financially attractive. Biomass waste-to-energy systems showed moderate sensitivity. Their financial performance is closely tied to the cost and availability of organic waste materials. While a 10% increase in equipment costs had little impact, a reduction in subsidies or a 15% increase in fuel-related logistics costs affected their viability more noticeably.

Overall, the analysis confirms that solar PV systems are the most financially robust, even when facing moderate increases in costs or changes in subsidies. Wind energy,

however, is highly sensitive to such changes, underlining the importance of sustained government support to make it a competitive option in Nigeria [71].

Life Cycle Assessment (LCA)

This section evaluates the long-term environmental impact of three key renewable energy technologies—solar PV, wind energy, and biomass—through a Life Cycle Assessment (LCA). The LCA tracks each system's energy use and emissions from production and installation to operation and decommissioning. The analysis focuses on two main indicators: the Energy Payback Period (EPP) and annual reductions in greenhouse gas (GHG) emissions [72].

The Energy Payback Period (EPP) measures how long it takes a system to generate the same amount of energy that was consumed during its production and installation. It is calculated as:

$$EPP = \frac{\text{Total Energy Used for Production}}{\text{Annual Energy Output}} \quad (1)$$

- a. Solar PV systems showed the shortest EPP, ranging from 2 to 3 years. With about 6,000 kWh/kW needed for production and annual outputs of 2,500–3,000 kWh/kW in Nigerian cities, solar PV becomes energy-positive quickly.
- b. Wind energy systems had an EPP of 5 to 6 years. With higher embodied energy (around 10,000–12,000 kWh/kW) and moderate output (~2,000 kWh/year), they take longer to offset their production energy.
- c. Biomass systems had a payback period of 4 to 5 years, depending on how efficiently the feedstock is collected and processed.
- d. GHG savings were calculated by comparing renewable generation to the emissions from grid electricity, using a Nigerian grid emission factor of 0.7 kg CO₂ per kWh. The formula used is:
- e. GHG Reduction (tons CO₂/year) = (Annual Output × 0.7) / 1,000
- f. Solar PV systems showed the highest reductions, achieving 50–70 tons of CO₂ savings per year for systems around 30 kW capacity.
- g. Biomass waste-to-energy systems followed, reducing 40–60 tons annually by displacing fossil fuel energy and avoiding methane emissions from waste.
- h. Wind energy systems reduced 30–50 tons of CO₂ annually depending on wind availability and location.

Among the three technologies, solar PV systems are the most environmentally efficient (Table 4). They provide the fastest energy payback and the greatest annual GHG reductions. Biomass systems offer strong environmental benefits by transforming organic waste into clean energy while offsetting methane emissions. Wind energy, while clean, has a longer payback period and is more site-dependent. These findings highlight the value of choosing technology based on both economic and environmental performance in urban planning [72, 73].

Table 2. Cost-Benefit Analysis for Renewable Energy Solutions

Renewable Energy Source	Capital Cost (USD/kW)	Payback Period (Years)	Net Present Value (NPV)
Solar Photovoltaic (PV)	1,200 – 1,800	6 – 8	Positive
Wind Energy (Small-scale)	2,000 – 2,500	10 – 12	Neutral
Biomass (Waste-to-Energy)	1,800 – 2,200	8 – 10	Positive

Table 3. Sensitivity Analysis – Impact of Variable Changes on Financial Viability

Variable	Solar PV	Wind Energy	Biomass
10% increase in equipment costs	Moderate	Significant	Low
10% decrease in government subsidy	Moderate	Significant	Moderate
15% increase in fuel costs	Low	Low	Moderate

Table 4. Life Cycle Assessment (LCA) of Renewable Energy Solutions

Renewable Energy Source	Energy Payback Period (Years)	GHG Emissions Reduction (tons CO ₂ /year)
Solar Photovoltaic (PV)	2 – 3	50 – 70
Wind Energy (Small-scale)	5 – 6	30 – 50
Biomass (Waste-to-Energy)	4 – 5	40 – 60

Challenges Identified

Despite the promising outcomes of renewable energy projects in Nigerian cities, several challenges still limit widespread adoption. The biggest barrier is the high upfront cost of installing systems, especially for wind and biomass technologies. Without government subsidies or financial support, these costs remain unaffordable for many homeowners and developers. Another issue is the lack of skilled local labor. Many installations require foreign technical support, which increases costs and slows down project delivery [74].

Furthermore, inconsistent enforcement of renewable energy policies, combined with the limited availability of financial incentives—such as tax breaks or subsidies—discourages private sector investment. While Nigeria has introduced renewable energy policies, weak enforcement and lack of clear implementation strategies continue to hinder progress [75].

Discussion

This section interprets the study's findings on energy savings, cost-effectiveness, environmental impacts, and policy gaps. It also compares the results with global literature and offers insight into the practical challenges and opportunities for scaling renewable energy solutions in Nigeria's urban buildings.

Energy Savings and Environmental Impact

Solar PV systems were the most effective in reducing electricity consumption, delivering savings of 25% to 35% across both residential and commercial buildings. These figures are consistent with global studies on solar PV, especially in regions with strong solar radiation like Nigeria [76, 77]. Energy savings in Lagos and Abuja were comparable to projects in other sub-Saharan African

countries, where similar systems report savings between 20% and 35% depending on the building type [78].

In terms of environmental impact, solar PV systems provided the highest reduction in greenhouse gas (GHG) emissions—50 to 70 tons of CO₂ annually per building. These outcomes match international findings that highlight solar energy as a leading solution for reducing urban carbon footprints [79]. The biomass waste-to-energy system in Port Harcourt also showed strong environmental benefits, cutting emissions by 40 to 60 tons of CO₂ annually. These systems help manage organic waste while also reducing methane emissions, which are especially harmful to the climate [80].

Life cycle assessments (LCA) confirmed solar PV's environmental advantages, with energy payback periods of just 2 to 3 years. In comparison, wind and biomass systems had longer payback periods of 4 to 6 years, making solar more appealing in terms of both energy and emissions performance [81].

Financial Viability of Renewable Systems

The cost-benefit analysis (CBA) highlighted solar PV as the most financially viable option. Projects in Lagos and Abuja showed a positive net present value (NPV) and a 6–8 year payback period. These findings are consistent with global trends where declining solar technology costs and favorable climate conditions have made solar PV cost-competitive [82, 83].

Biomass waste-to-energy systems also proved financially viable, with an 8–10 year payback period. This was largely due to the low cost of feedstock and income from electricity generation. These results support previous studies showing that biomass systems work well in regions with abundant organic waste [84].

In contrast, wind energy systems faced financial hurdles. Their payback periods were longer—10 to 12 years—due to higher capital costs and less favorable wind conditions in many Nigerian cities. Without financial support, these systems are not competitive compared to solar or biomass [85, 86].

Sensitivity to Costs and the Importance of Policy

Sensitivity analysis revealed that solar PV systems remained financially sound even when equipment costs increased by 10%. This shows the robustness of solar investments, especially in sun-rich regions like Nigeria [85]. However, wind energy projects were highly sensitive to cost changes and depended heavily on policy incentives to remain viable. Without strong government support, wind energy remains a risky investment [86].

Biomass systems showed moderate sensitivity to fluctuations in feedstock supply and cost. Their success depends on stable waste collection systems and steady energy demand, particularly from industrial users. In places like Port Harcourt—where waste disposal is a major issue—biomass offers an effective solution but requires coordination between stakeholders [87].

Overall, the analysis shows that strong policy support is essential for renewable energy adoption. Without financial incentives—such as subsidies, tax relief, or low-interest loans—costlier technologies like wind and biomass are unlikely to scale [88]. The success of solar PV systems across Nigeria demonstrates the impact well-designed policies and incentives can have on accelerating renewable energy deployment [84].

Key Barriers and Opportunities for Scale-Up

Several obstacles continue to restrict renewable energy growth in Nigeria's cities. High capital costs remain a major challenge, especially for wind and biomass systems. Even though solar PV has become more affordable, many small businesses and households still cannot afford the upfront costs without financing options [78, 82, 89].

Another key challenge is the lack of technical skills within Nigeria. Complex systems often rely on foreign technicians, which raises costs and slows adoption. Building a local workforce through technical education and vocational training is essential for lowering project costs and scaling the industry [89].

Policy enforcement is also a major gap. Although Nigeria has solid renewable energy frameworks—such as the Renewable Energy Master Plan (REMP) and the National Renewable Energy and Energy Efficiency Policy (NREEEP)—poor enforcement has weakened their impact. Research from other developing countries also shows that policy inconsistency is a major barrier to renewable energy adoption [86, 90].

Policy Implications and Recommendations

Nigeria's current renewable energy policies—particularly the Renewable Energy Master Plan (REMP) and the National Renewable Energy and Energy Efficiency Policy

(NREEEP)—outline ambitious targets for solar, wind, biomass, and hydropower adoption. However, weak enforcement and lack of local adaptation have limited their effectiveness [90]. Policy implementation varies by city: Lagos benefits from local solar initiatives, Abuja has gained from federal housing solar programs, while Port Harcourt lags due to industrial fossil fuel dominance and minimal incentive application.

To accelerate renewable energy adoption across urban centers, the following targeted recommendations are proposed:

- a. **Expand Financial Incentives:** Government-backed subsidies, tax reliefs, and low-interest loans should be scaled up to address high upfront costs. Tailored incentives should support rooftop solar in Lagos, residential solar in Abuja, and biomass installations in Port Harcourt's industrial zones. This approach mirrors effective models from India and South Africa where incentives drove market growth [90].
- b. **Strengthen and Localize Policy Enforcement:** Existing policies must be made legally enforceable and context-sensitive. In Lagos, local building codes should require renewable integration in commercial structures. Abuja can introduce mandatory energy standards for new residential estates, while Port Harcourt should link biomass adoption to environmental compliance for industries [91].
- c. **Build Local Technical Capacity:** Investments in technical training will reduce reliance on foreign expertise [89]. In Lagos, training should focus on PV systems and battery storage; in Abuja, on off-grid and hybrid systems; and in Port Harcourt, on biomass energy for waste-to-power applications. Public-private partnerships, as demonstrated in Kenya and Ethiopia, can support curriculum development and workforce training [92, 93].
- d. **Promote Public-Private Partnerships (PPPs):** PPPs can mobilize capital and expertise for urban energy projects. In Lagos, this can include solar for public institutions; in Abuja, hybrid systems for government and housing developments; and in Port Harcourt, biomass solutions for industrial parks. Frameworks similar to South Africa's REIPPPP can help de-risk investments [94].

In sum, while solar PV is already proving viable in Nigeria's urban centers, scaling up renewable energy adoption—especially for wind and biomass—requires strong, localized policy support, financial mechanisms, technical capacity building, and collaboration between the public and private sectors.

5. CONCLUSION

This study has shown the strong potential of renewable energy solutions—particularly solar photovoltaic (PV) systems—for addressing the rising energy demands in Nigeria's urban centers: Lagos, Abuja, and Port Harcourt. Among the technologies assessed, solar PV emerged as

the most effective, offering energy savings of 25% to 35% and a relatively short payback period of 6 to 8 years. These systems reduce reliance on the national grid while delivering environmental benefits, including annual greenhouse gas (GHG) emissions reductions of 50 to 70 tons of CO₂.

Biomass waste-to-energy systems also demonstrated promise, especially in industrial contexts, achieving energy savings of up to 20% and contributing to emission reductions. However, wind energy remains the least viable option due to its high capital cost and extended payback period, making it less attractive without government support.

The cost-benefit analysis confirmed the financial viability of solar PV and biomass systems for urban buildings, while the sensitivity analysis revealed that solar PV maintains strong financial performance even under fluctuating equipment costs and shifting policy conditions. Wind energy, by contrast, proved highly sensitive to such variables, underscoring the need for targeted subsidies and policy incentives to improve its feasibility. Life cycle assessment results further emphasized the environmental advantages of solar PV, with short energy payback times and substantial long-term emissions reductions.

Despite these positive findings, several challenges continue to slow the adoption of renewable energy in Nigeria. Chief among these are high upfront costs—especially for wind and biomass—alongside a shortage of skilled local professionals, which increases the cost of installation and system maintenance. Inconsistent policy enforcement and a lack of robust financial incentives further limit the expansion of renewable energy technologies.

To address these barriers, this study recommends a range of strategic policy actions. First, the Nigerian government should enhance financial incentives, including subsidies, tax relief, and low-interest loans, to encourage private sector investment. Second, better enforcement of existing renewable energy policies, such as the Renewable Energy Master Plan (REMP) and the National Renewable Energy and Energy Efficiency Policy (NREEEP), is essential to establish a stable and supportive regulatory environment. Third, investment in technical training and education programs is critical to build local capacity and reduce dependence on foreign expertise. Finally, fostering public-private partnerships (PPPs) can help mobilize funding and accelerate the implementation of renewable energy projects at scale.

Overall, the integration of renewable energy technologies into Nigeria's urban buildings presents a powerful opportunity to cut energy costs, improve energy security, and reduce environmental impacts. With the right mix of policy support, financial tools, and capacity-building initiatives, renewable energy can serve as a cornerstone of sustainable urban development in Nigeria. Future research should focus on gathering real-time performance data from active projects and exploring hybrid systems that

combine solar, wind, and biomass to meet the diverse energy needs of urban areas.

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